

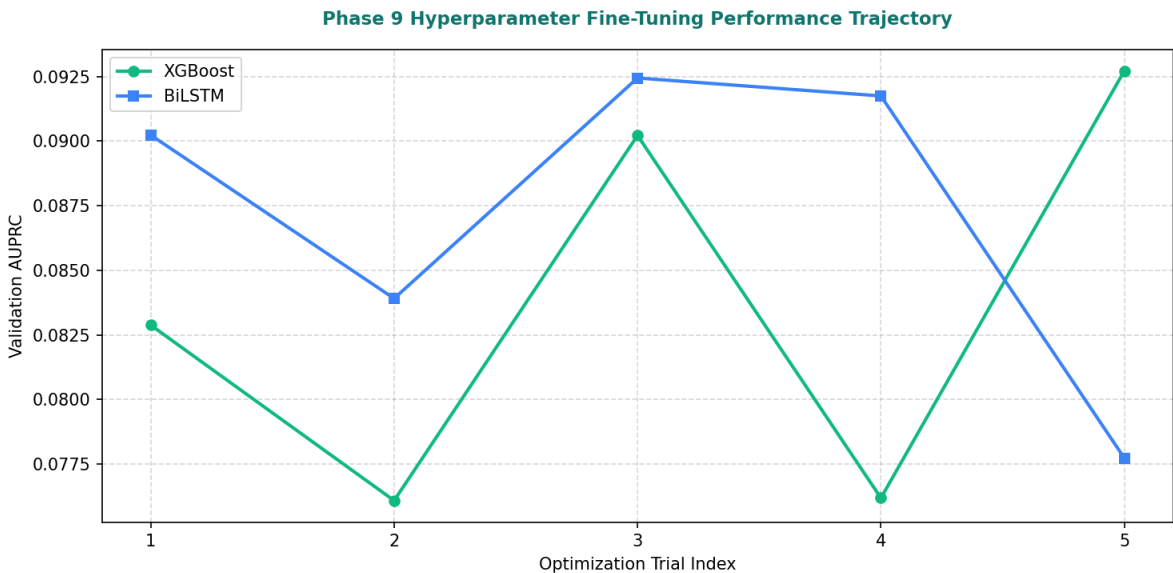
THAARU Sepsis AI — Phase 9 Optimization Report

Hyperparameter Fine-Tuning Trajectories • Generated: July 19, 2026

1. Executive Summary

This report documents Phase 9: Hyperparameter Optimization. We ran randomized sweeps to tune XGBoost and BiLSTM classifiers to maximize validation AUPRC under severe imbalance. Fine-tuning details are logged below.

2. Comparative Performance Plot



3. Optimized Results Recommendations

Best XGBoost Model: AUPRC: 0.0927 (depth=4, lr=0.1, n_estimators=100)

Best BiLSTM Model: AUPRC: 0.0924 (hidden=32, layers=1, dropout=0.3)